Propositional Logic

Birzhan Kalmurzayev

Ac. year 2025-2026

Boolean functions

Definition

Any function $f : \{0,1\}^n \to \{0,1\}$ (with inputs and output are only 0 or 1) is called Boolean function.

Boolean functions

Definition

Any function $f : \{0, 1\}^n \to \{0, 1\}$ (with inputs and output are only 0 or 1) is called Boolean function.

Logical functions can be represented as Boolean functions, if

- 1 is mean true statement, and
- 0 is mean false statement

Boolean functions

Definition

Any function $f : \{0,1\}^n \to \{0,1\}$ (with inputs and output are only 0 or 1) is called Boolean function.

Logical functions can be represented as Boolean functions, if

- 1 is mean true statement, and
- 0 is mean false statement

How many Boolean functions with n variable can have?

Boolean functions

Definition

Any function $f : \{0,1\}^n \to \{0,1\}$ (with inputs and output are only 0 or 1) is called Boolean function.

Logical functions can be represented as Boolean functions, if

- 1 is mean true statement, and
- 0 is mean false statement

How many Boolean functions with n variable can have?
Answer: 2^{2^n}.

Boolean functions with 1 variable

The following is all Boolean function with one variable:

- $f_1(0) = 0,\ f_1(1) = 0;$
- $f_2(0) = 0,\ f_2(1) = 1;$
- $f_3(0) = 1,\ f_3(1) = 0;$
- $f_4(0) = 1,\ f_4(1) = 1;$

Boolean functions with 1 variable

The following is all Boolean function with one variable:

- $f_1(0) = 0, \; f_1(1) = 0;$
- $f_2(0) = 0, \; f_2(1) = 1;$
- $f_3(0) = 1, \; f_3(1) = 0;$
- $f_4(0) = 1, \; f_4(1) = 1;$

These functions are conveniently written as the following table (We will call it Truth table)

x	f_1	f_2	f_3	f_4
0	$f_1(0)$	$f_2(0)$	$f_3(0)$	$f_4(0)$
1	$f_1(1)$	$f_2(1)$	$f_3(1)$	$f_4(1)$

Boolean functions with 1 variable

The following is all Boolean function with one variable:

- $f_1(0) = 0, \ f_1(1) = 0;$
- $f_2(0) = 0, \ f_2(1) = 1;$
- $f_3(0) = 1, \ f_3(1) = 0;$
- $f_4(0) = 1, \ f_4(1) = 1;$

These functions are conveniently written as the following table (We will call it Truth table)

x	f_1	f_2	f_3	f_4
0	0	0	1	1
1	0	1	0	1

Boolean functions with 1 variable

As a logical function

- f_1 corresponds to the false function, lets denote it as 0;
- f_2 corresponds to the identity ($f(x) = x$), lets denote it as x;
- f_3 corresponds to the negation of x, lets denote it as $\bar{x}$;
- f_4 corresponds to the true function, lest denote it as 1.

These functions are conveniently written as the following table (We will call it Truth table)

x	0	x	$\bar{x}$	1
0	0	0	1	1
1	0	1	0	1

Boolean functions with 2 variables

Truth table of a Boolean function ($f(x, y)$) with 2 variables will consist 4 rows

- fist row for value when $x = 0$ and $y = 0$;
- second row for value when $x = 0$ and $y = 1$;
- third row for value when $x = 1$ and $y = 0$;
- fourth row for value when $x = 1$ and $y = 1$.

Boolean functions with 2 variables

Truth table of a Boolean function ($f(x, y)$) with 2 variables will consist 4 rows

- fist row for value when $x = 0$ and $y = 0$;
- second row for value when $x = 0$ and $y = 1$;
- third row for value when $x = 1$ and $y = 0$;
- fourth row for value when $x = 1$ and $y = 1$.

x	y																
0	0																
0	1																
1	0																
1	1																

Boolean functions with 2 variables

Truth table of a Boolean function ($f(x, y)$) with 2 variables will consist 4 rows

- fist row for value when $x = 0$ and $y = 0$;
- second row for value when $x = 0$ and $y = 1$;
- third row for value when $x = 1$ and $y = 0$;
- fourth row for value when $x = 1$ and $y = 1$.

x	y																
0	0	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1	1
0	1	0	0	0	0	1	1	1	1	0	0	0	0	1	1	1	1
1	0	0	0	1	1	0	0	1	1	0	0	1	1	0	0	1	1
1	1	0	1	0	1	0	1	0	1	0	1	0	1	0	1	0	1

Boolean functions with 2 variables

Truth table of a Boolean function ($f(x, y)$) with 2 variables will consist 4 rows

- fist row for value when $x = 0$ and $y = 0$;
- second row for value when $x = 0$ and $y = 1$;
- third row for value when $x = 1$ and $y = 0$;
- fourth row for value when $x = 1$ and $y = 1$.

x	y	0			x		y					$\bar{y}$		$\bar{x}$			1
0	0	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1	1
0	1	0	0	0	0	1	1	1	1	0	0	0	0	1	1	1	1
1	0	0	0	1	1	0	0	1	1	0	0	1	1	0	0	1	1
1	1	0	1	0	1	0	1	0	1	0	1	0	1	0	1	0	1

Boolean function with 2 variables

A new names for Boolean functions with 2 variables:

- $x \cdot y$ – is conjunction or multiplication of x and y;
- $x \vee y$ – is disjunction of x and y;
- $x + y$ – is sum by modulo 2;
- $x \to y$ – is implication (x implies y);
- $x \sim y$ – is equivalence of x and y;
- $x|y$ – Sheffer strough (anticonjunction of x and y);
- $x \downarrow y$ – Peirce arrow (antidusjunction of x and y).

Boolean functions with 2 variables

x	y	0	$x \cdot y$	$\overline{x \to y}$	x	$\overline{y \to x}$	y	$x + y$	$x \vee y$
0	0	0	0	0	0	0	0	0	0
0	1	0	0	0	0	1	1	1	1
1	0	0	0	1	1	0	0	1	1
1	1	0	1	0	1	0	1	0	1

x	y	$x \downarrow y$	$x \sim y$	$\bar{y}$	$y \to x$	$\bar{x}$	$x \to y$	$x \mid y$	1
0	0	1	1	1	1	1	1	1	1
0	1	0	0	0	0	1	1	1	1
1	0	0	0	1	1	0	0	1	1
1	1	0	1	0	1	0	1	0	1

Vector Form of a Boolean Function

Definition

Let $f : \{0,1\}^n \to \{0,1\}$ be a Boolean function. Fix an ordering of inputs $\alpha_1, \alpha_2, \ldots, \alpha_{2^n} \in \{0,1\}^n$ (e.g., lexicographic order). The **vector form** of f is the binary vector

$$\mathbf{v}(f) = \big(f(\alpha_1)\ f(\alpha_2)\ \ldots\ f(\alpha_{2^n})\big) \in \{0,1\}^{2^n}.$$

Example.
For $n = 2$ (order $00, 01, 10, 11$), if

$$f(x, y) = x + y,$$

then

$$\mathbf{v}(f) = (0110).$$

Example: If vector of the functions f and g are $v(f) = (0001)$ and $v(g) = (1101)$, then find the vector of function

$$h(x, y, z) = f(y, g(x, z)) \vee g(y, z)$$

Example: If vector of the functions f and g are $v(f) = (0001)$ and $v(g) = (1101)$, then find the vector of function

$$h(x, y, z) = f(y, g(x, z)) \vee g(y, z)$$

Build Truth Table for the function

$$((x|y)|z) \to x$$

The main equivalences

Definition

Two Boolean functions f and g is called equivalence (or equal, and we denote $f = g$) if they have same values for any inputs (or same Truth table).

The main equivalences

Definition

Two Boolean functions f and g is called equivalence (or equal, and we denote $f = g$) if they have same values for any inputs (or same Truth table).

The Main Equivalences:

1. All the following operations are commutative $\{\cdot;\ \vee;\ +;\ \sim;\ |;\ \downarrow\}$
2. All the following operations are associative $\{\cdot;\ \vee;\ +;\ \sim\}$;
3. Distributive laws:
   - $(x \vee y) \cdot z = (x \cdot z) \vee (y \cdot z)$;
   - $(x \cdot y) \vee z = (x \vee z) \cdot (y \vee z)$;
   - $(x + y) \cdot z = (x \cdot z) + (y \cdot z)$.

The main equivalences

4. De Morgan laws:
   - $\overline{x \cdot y} = \bar{x} \vee \bar{y}$;
   - $\overline{x \vee y} = \bar{x} \cdot \bar{y}$.
5. Absorption laws:
   - $x \vee (x \cdot y) = x$;
   - $x \cdot (x \vee y) = x$.
6. $x \to y = \bar{x} \vee y$
7. $x \sim y = \overline{x + y} = (x \cdot y) \vee (\bar{x} \cdot \bar{y}) = (x \vee \bar{y}) \cdot (\bar{x} \vee y)$;
8. $x|y = \overline{x \cdot y} = \bar{x} \vee \bar{y}$;
9. $x \downarrow y = \overline{x \vee y} = \bar{x} \cdot \bar{y}$

Example. Show that the following functions are equivalent

$$f(x,y,z) = \overline{(z|y)}|((z \downarrow x) \vee \bar{z}), \quad g(x,y,z) = (x \downarrow (y \to x)) \to (y \vee z)$$

Functionally Complete Sets of Boolean Functions

Definition

A set $\mathcal{F}$ of Boolean functions is called **functionally complete** (or **complete**) if every Boolean function can be obtained from functions in $\mathcal{F}$ by superposition.

Examples.

1. $\{\bar{\ }, \cdot\}$ is complete.
2. $\{\bar{\ }, \vee\}$ is complete.
3. $\{|\}$ (NAND) is complete.
4. $\{\downarrow\}$ (NOR) is complete.
5. $\{\cdot, +\}$

DNF and CNF of Boolean Functions

Definition

If x is a variable, then both x and $\bar{x}$ we call **literal**.

A Boolean formula is in **DNF (Disjunctive Normal Form)**, if it has form

$$K_1 \vee K_2 \vee \cdots \vee K_m,$$

where each K_i is a conjunction of literals.

A Boolean formula is in **CNF (Conjunctive Normal Form)**, if it has form

$$K_1 \wedge K_2 \wedge \cdots \wedge K_m,$$

where each K_i is a disjunction of literals.

Perfect (Full) DNF and CNF

Definition

Perfect Disjunctive Normal Form (PDNF).
A DNF is called **perfect** (or **full**) if

- each conjunction (minterm) contains *all* variables,
- each variable appears exactly once (either x_i or $\overline{x_i}$).

Perfect Conjunctive Normal Form (PCNF).
A CNF is called **perfect** if

- each disjunction (maxterm) contains *all* variables,
- each variable appears exactly once.

Theorem

Every Boolean function has a unique PDNF and PCNF (up to reordering of terms).

Theorem

Every Boolean function has a unique PDNF and PCNF (up to reordering of terms).

Example. Find DNF, CNF, PDNF and PCNF of the following function

$$(x \vee y) | \overline{(z \sim \bar{x})}$$

Precedence of Boolean Operations

When parentheses are omitted, Boolean operations are evaluated according to the following priority:

1. $\bar{x}$
2. $x \cdot y$
3. $x \vee y$
4. $x + y$
5. $x \to y$, $x \sim y$

Precedence of Boolean Operations

When parentheses are omitted, Boolean operations are evaluated according to the following priority:

1. $\bar{x}$
2. $x \cdot y$
3. $x \vee y$
4. $x + y$
5. $x \rightarrow y$, $x \sim y$

Example.

$$\bar{x} \vee y \cdot z = \bar{x} \vee (y \cdot z)$$

$$x \rightarrow y \vee z = x \rightarrow (y \vee z)$$

Zhegalkin Polynomial (Algebraic Normal Form)

Theorem

Every Boolean function $f : \{0,1\}^n \to \{0,1\}$ can be uniquely represented in the form

$$f(x_1, \ldots, x_n) = a_0 + \sum_i a_i x_i + \sum_{i<j} a_{ij} x_i x_j + \cdots + a_{12\ldots n} x_1 x_2 \ldots x_n,$$

where

$$a_0, a_i, a_{ij}, \cdots \in \{0,1\},$$

This representation is called the **Zhegalkin polynomial** (or **Algebraic Normal Form, ANF**).

Computing Zhegalkin Coefficients (Triangle Method)

Idea. Write the vector of values of f in lexicographic order. Construct a triangular table by taking pairwise sums modulo 2.

Example. Let $n = 2$, order $00, 01, 10, 11$, and $\mathbf{v}(f) = (0, 1, 1, 0)$.

$$\begin{array}{cccc} \mathbf{0} & 1 & 1 & 0 \\ & \mathbf{1} & 0 & 1 \\ & & \mathbf{1} & 1 \\ & & & \mathbf{0} \end{array} \qquad \text{(all sums mod 2)}$$

Rule. The leftmost elements of each row are the coefficients of the Zhegalkin polynomial (ANF): $a_0,\ a_1,\ a_2,\ a_{12}$.
Result. Here we obtain $f(x_1, x_2) = x_1 + x_2$.

Example. Find Zhegalkin polynomial of the following function

$$x \to ((y \vee z) \downarrow x)$$